

3.3 The Test for Divergence & Other Theorems

Theorem: If the series $\sum_{i=1}^{\infty} a_i$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$.

Proof:

Let s_n denote the n^{th} partial sum of the series $\sum_{i=1}^{\infty} a_i$. For $n \geq 2$, we have $s_n - s_{n-1} = a_n$. Since the series is convergent, we know that there is some number, s , such that $\lim_{n \rightarrow \infty} s_n = s$. Since $n-1 \rightarrow \infty$ as $n \rightarrow \infty$, we have that $\lim_{n \rightarrow \infty} s_{n-1} = s$. Thus, the sequence given by $\{s_n - s_{n-1}\}$ is convergent and by the limit laws:

$$s_n - s_{n-1} = a_n \Rightarrow \lim_{n \rightarrow \infty} (s_n - s_{n-1}) = \lim_{n \rightarrow \infty} a_n \Rightarrow 0 = \lim_{n \rightarrow \infty} a_n.$$

Note: The above proof illustrates the importance of understanding the two sequences associated with a series. Namely the sequence of its partial sums, $\{s_n\}$, and the sequence of its terms, $\{a_n\}$.

Corollary (Test for Divergence): If $\lim_{n \rightarrow \infty} a_n$ does not exist or if $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series $\sum_{i=1}^{\infty} a_i$ is divergent.

Proof:

This is simply the contrapositive of the above theorem.

Example 1: Prove that the series $\sum_{n=1}^{\infty} \left(\frac{n^2}{5n^2 + 4} \right)$ diverges.

We begin by examining $\lim_{n \rightarrow \infty} a_n$ where $a_n = \frac{n^2}{5n^2 + 4}$. Using the ideas of section 5.1.1 we consider the continuous

function $f(x) = \frac{x^2}{5x^2 + 4}$. We note that by l'Hospital's rule $\lim_{x \rightarrow \infty} \frac{x^2}{5x^2 + 4} = \lim_{x \rightarrow \infty} \frac{2x}{10x} = \lim_{x \rightarrow \infty} \frac{1}{5} = \frac{1}{5}$. So, we know

$\lim_{n \rightarrow \infty} a_n = \frac{1}{5} \neq 0$. So, this series diverges by the test for divergence.

Example 2 (you try): In no more than two sentences explain what the test for divergence tells you about the series $\sum_{k=1}^{\infty} \left(\frac{1}{3k} \right)$.

We first note that if $a_n = \frac{1}{3n}$, then $\lim_{n \rightarrow \infty} a_n = \frac{1}{3} \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right) = 0$. Thus, the test for divergence tells us nothing about whether this series converges or diverges.

Theorem: If $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent series, then so are the series $\sum_{n=1}^{\infty} ca_n$ (where c is a constant), $\sum_{n=1}^{\infty} (a_n + b_n)$, and $\sum_{n=1}^{\infty} (a_n - b_n)$, and

- $\sum_{n=1}^{\infty} (a_n \pm b_n) = \sum_{n=1}^{\infty} a_n \pm \sum_{n=1}^{\infty} b_n$
- $\sum_{n=1}^{\infty} ca_n = c \sum_{n=1}^{\infty} a_n$

Proof: The proof follows from the limit laws of section 3.1.1.

Example 3: Determine (with justification) whether the series is convergent or divergent. If it is convergent, find its sum: $\frac{1}{3} + \frac{2}{7} + \frac{1}{27} + \frac{2}{49} + \frac{1}{243} + \frac{2}{343} + \dots$.

We first note that we can rewrite the series as:

$$\begin{aligned} & \frac{1}{3} + \frac{2}{7} + \frac{1}{27} + \frac{2}{49} + \frac{1}{243} + \frac{2}{343} + \dots \\ &= \left(\frac{1}{3} + \frac{2}{7} \right) + \left(\frac{1}{27} + \frac{2}{49} \right) + \left(\frac{1}{243} + \frac{2}{343} \right) + \dots \\ &= \sum_{k=1}^{\infty} \left(\frac{1}{3} \left(\frac{1}{9} \right)^{k-1} + \frac{2}{7} \left(\frac{1}{7} \right)^{k-1} \right). \end{aligned}$$

We recognize $\sum_{k=1}^{\infty} \left(\frac{1}{3} \left(\frac{1}{9} \right)^{k-1} \right)$ and $\sum_{k=1}^{\infty} \left(\frac{2}{7} \left(\frac{1}{7} \right)^{k-1} \right)$ as geometric series with common ratio less than one. So, both of the series are convergent and by the above theorem the series in question is convergent and:

$$\sum_{k=1}^{\infty} \left(\frac{1}{3} \left(\frac{1}{9} \right)^{k-1} + \frac{2}{7} \left(\frac{1}{7} \right)^{k-1} \right) = \sum_{k=1}^{\infty} \left(\frac{1}{3} \left(\frac{1}{9} \right)^{k-1} \right) + \sum_{k=1}^{\infty} \left(\frac{2}{7} \left(\frac{1}{7} \right)^{k-1} \right) = \frac{\left(\frac{1}{3} \right)}{\left(1 - \frac{1}{9} \right)} + \frac{\left(\frac{2}{7} \right)}{\left(1 - \frac{1}{7} \right)} = \frac{\left(\frac{1}{3} \right)}{\left(\frac{8}{9} \right)} + \frac{\left(\frac{2}{7} \right)}{\left(\frac{6}{7} \right)} = \frac{3}{8} + \frac{1}{3} = \frac{17}{24}.$$

■

Theorem: Convergence or divergence is unaffected by deleting a finite number of terms from a series.
That is, for any positive integer K , the series:

$$\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \dots$$

$$\sum_{k=K}^{\infty} a_k = a_K + a_{K+1} + a_{K+2} + \dots$$

both converge or both diverge.

Example 4: Determine whether the following series converges or diverges. $\sum_{i=10}^{\infty} \frac{1}{i} = \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \dots$

We can see that $\sum_{i=10}^{\infty} \frac{1}{i}$ looks like the harmonic series $\sum_{i=1}^{\infty} \frac{1}{i}$ with the first nine terms deleted. Therefore since the harmonic series diverges the given series $\sum_{i=10}^{\infty} \frac{1}{i}$ diverges. ■

Example 5 (you try):

(a) Determine (with justification) whether the series is convergent or divergent. If it is convergent, find its sum: $\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right)$.

We need to examine the two series: $\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} \right)$ and $\sum_{n=1}^{\infty} \left(\frac{1}{2^n} \right)$. The latter series is convergent since it is a

geometric series with common ratio $\frac{1}{2} < 1$. Moreover, from the previous lecture we know $\sum_{n=1}^{\infty} \left(\frac{1}{2^n} \right) = \frac{\left(\frac{1}{2} \right)}{\left(1 - \frac{1}{2} \right)} = 1$

. For the first series we notice that $\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} \right) = \sum_{n=1}^{\infty} 3 \left(\frac{1}{n(n+1)} \right)$. We also know from yesterday's lecture

(using partial fraction decomposition) that $\sum_{n=1}^{\infty} \left(\frac{1}{n(n+1)} \right) = 1$, thus we know that $\sum_{n=1}^{\infty} 3 \left(\frac{1}{n(n+1)} \right)$ is convergent

and $\sum_{n=1}^{\infty} 3 \left(\frac{1}{n(n+1)} \right) = 3 \sum_{n=1}^{\infty} \left(\frac{1}{n(n+1)} \right) = 3$. Therefore, we know that $\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right)$ is convergent and

$$\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right) = \sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{2^n} \right) = 3 + 1 = 4.$$

■

- (b) Determine (with justification) whether the series is convergent or divergent. If it is convergent, find its sum: $\sum_{n=1}^{\infty} (\arctan(-n))$.

We begin by examining $\lim_{n \rightarrow \infty} a_n$ where $a_n = \arctan(-n)$. Using the ideas of section 5.1.1 we consider the

function $f(x) = \arctan(-x)$. From calculus 1 we know $\lim_{x \rightarrow \infty} \arctan(-x) = \frac{-\pi}{2}$. So, we know that

$\lim_{n \rightarrow \infty} a_n = \frac{-\pi}{2} \neq 0$. So, this series diverges by the test for divergence.

■

- (c) TRUE or FALSE: If $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are divergent series, then $\sum_{n=1}^{\infty} (a_n + b_n)$ is a divergent series. If you answer TRUE you must explain why this is TRUE. If you answer FALSE you must provide a counterexample proving the statement FALSE.

This is FALSE. Consider the two series $\sum_{k=1}^{\infty} \frac{1}{k}$ and $\sum_{k=1}^{\infty} \frac{-1}{k}$. They are clearly both divergent by what we showed

in 3.2. However, $\sum_{k=1}^{\infty} \left(\frac{1}{k} + \frac{-1}{k} \right) = \sum_{k=1}^{\infty} (0) = 0$ is convergent. Thus, the statement is FALSE.